Immobilization of cane amylase and acid phosphatase on tricalcium phosphate (TCP) gel

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Amylase (E.C. 3.2.1.1) and acid phosphatase (E.C. 3.1.3.2) from sugar cane juice have been isolated and purified by salt and solvent fractionation. They have been separated by ion-exchange chromatography using DEAE cellulose and phosphocellulose columns and were immobilized on tricalcium phosphate gel. Some of the properties of immobilized and free soluble enzymes have been compared. The pH and temperature profiles of free and immobilized amylase and acid phosphatase did not show much alteration. Thermal stability of both the enzymes increased in the presence of 18% sucrose solution. The K_m *value of amylase and acid phosphatase increased on immobilization, while the* V_{max} *values showed wide variation. Application of co-immobilized amylase and acid phosphatase in the removal of starch from cane juice has been suggested.*

Keywords: Immobilization; amylase; phosphatase; sugar cane; tricalcium phosphate

Introduction

There has been great interest in the study of co-immobilized enzymes over the last few years for different uses.[1–4] It is believed that these enzymes will find increasing application in industrial, analytical, and therapeutic operations. The high cost and unstable nature of enzymes have restricted the industrial use of enzymes. Cane amylase plays a great role in the metabolism of sugar cane plant.[5]

Alexander[6] has purified amylase from sugar cane leaves by gel filtration technique. The bacterial and cane amylase (E.C. 3.2.1.1) have been reported to remove the starch from cane juice and improve filtration.[7–8] The application of cane acid phosphatase (E.C. 3.1.3.2) in the release of inorganic phosphorus and cane amylase for the removal of starch from cane juice for better clarification has been suggested by the authors.[9,10]

The present study describes the purification of cane amylase and acid phosphatase and co-immobilization of partially purified enzymes on inert supports by different techniques.

In our present studies, the aim was to employ enzymes for clarification of cane juice, and thus the partially purified ammonium sulfate fractionated enzyme was taken up for co-immobilization on various supports such as Kieselguhr, Bentonite, Kaolin, Superphosphate, Alumina, etc., which are commonly employed as filter aids,[11] for clarification of juice in the sugar industry. Some of the parameters that affect the working of immobilized enzyme systems were studied and reported here.

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Materials and methods

Bentonite and DEAE cellulose (Sigma), iron oxide *p*-nitrophenyl phosphate (p-NPP), and 3,5 dinitrosalicylic acid (DNS, E. Merck), starch (potato), β-glycerophosphate, acrylamide, *N-N″* bis acrylamide (B.D.H.) celite (Koch Lite), and other chemicals used were of A.R. or equivalent grade.

Tricalcium phosphate (TCP) gel was prepared by following the method of Keilin *et al.*[12] and hydroxylapatite by the Jenkins method.[13] Magnetite was prepared by the method of Mark *et al.*[14]

Enzyme preparation and purification

Sugar cane juice (1000 ml) was filtered through muslin cloth and centrifuged at 3000 rev min^{-1} for 10 min in a refrigerated centrifuge. The clear supernatant was 30% saturated with ammonium sulfate, and the precipitate obtained (containing minor amylase activity) on centrifugation in cold (4°C) was discarded. The supernatant was raised to 80% saturation with further addition of ammonium sulfate and kept overnight in the refrigerator. The fractionated ammonium sulfate precipitate was removed by centrifuging in cold (4°C), dissolved

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in 0.1 M acetate buffer (pH 5.2), and dialyzed against distilled water at 4°C in the refrigerator for 24 h.

The dialyzed enzymes from ammonium sulfate fractionation obtained as above were treated with 10 mM of NaF at 30°C for 1 h and then reprecipitated with cold acetone to give 50% acetone solution.

The precipitate from the acetone fraction was dissolved in 0.1 M acetate buffer pH 5.2 and dialyzed as described above. The contents were analyzed for protein and enzyme activities at each stage.

The dialyzed enzyme solution (12 ml) was subjected to ion exchange chromatography on DEAE-cellulose column (1.5 × 10 cm) pre-equilibrated with 0.1 M acetate buffer pH 5.2 by following the Fleming and Stone procedure.[15] The column was eluted with 0.1 M acetate buffer (pH 5.4) containing 0.05 to 0.3 M NaCl at a flow rate of 1 to 1.5 ml min^{-1} at 20°C. Five milliliters of eluate portions collected at regular intervals were taken up for determination of specific activities of amylase and acid phosphatase. Peak activity fractions were combined and concentrated by adding 1.5 times the volume of cold acetone.

There was overlapping of acid phosphatase activity in amylase fractions, and for separating them a phosphocellulose column was used.

The active protein fraction from acetone was separated by centrifugation, dissolved in 0.1 M acetate buffer (pH 5.2), and dialyzed. The dialyzed enzyme was treated on phosphocellulose column (1.5 × 3.5 cm) as per the procedure described by Abe *et al.*[16] It was eluted with 0.1 M acetate buffer pH 5.4 containing 0.05 to 0.5 M NaCl. Amylase was found eluted with buffer containing 0.15 M NaCl, whereas acid phosphatase was eluted with buffer containing 0.3 M NaCl. The peak activities of enzyme fractions in each case containing these separate enzymes were combined and concentrated by adding cold acetone as described before and preserved in the refrigerator.

The partially purified dialyzed preparation obtained by ammonium sulfate fractionation having specific activity 17 for amylase and 0.18 for acid phosphatase was used in immobilization studies.

Enzyme assay

Amylase[17] and phosphatase[18] activities were tested using standard procedures.

Amylase. In the amylase assay, enzyme reaction mixtures containing 2.5 ml of 2% starch (potato), 2 ml of 0.1 M acetate buffer (pH 5.5), and 0.5 ml of enzyme 60–70 μg protein in a total volume of 5.0 ml was incubated at 50°C for 30 min. The reaction was stopped by adding 1 ml 0.5 M NaOH and reducing sugars released were estimated by the 3 : 5 dinitro salicylic acid (DNS) method.[19]

Phosphatase. The activity of acid phosphatase was assayed by incubating a reaction mixture of 3.0 ml containing 2.0 ml of 0.2 M acetate buffer, pH 5.2, 0.5 ml of enzyme (320 μg protein), and 0.5 ml of substrate (1% *p*-NPP/β-glycerophosphate) at 40°C for 60 min. The reaction was stopped by adding 1.0 ml of molybdate reagent and the inorganic phosphorus (p_i) released was estimated by the Fiske Subbarow method.[20] Protein was determined by Lowry's method.[21] One unit of enzyme activity (amylase/phosphatase) was defined as that quantity of enzyme which liberated 1.0 mg of reducing sugar/inorganic phosphorus in 1 h under the above assay conditions, and the specific activity was expressed as activity per milligram of protein.

Immobilization of enzymes by entrapment method

Calcium alginate beads.[22] Three percent sodium alginate solution (150 mg in 5 ml) was mixed with 6 mg of enzyme protein in a total volume of 5 ml of 0.1 M tris maleate buffer, pH 5.2. This was taken up in a syringe and was added slowly dropwise in 60 mM $CaCl_2$ solution to form uniform porous beads of approximately 3 mm diameter.

Iron oxide/magnetite beads.[22] Three percent alginate with 110 mg of iron oxide/magnetite[14] and 6 mg enzyme protein in (0.1 M) tris maleate buffer, pH 5.2, was mixed well by stirring (without entrapping air bubbles) and taken up in a syringe. Beads were prepared as above using $CaCl_2$ (60 mM).

Polyacrylamide gel.[23] Seven hundred fifty milligrams of acrylamide, 1.0 ml enzyme (6 mg protein), 41 mg of *N-N'* bis-acrylamide, 1.5 ml 0.1 M tris maleate buffer, pH 5.2, 1.5 ml 0.5% dimethylaminopropionitrile, and 1.5 ml 2.5% ammonium persulfate were mixed well and kept at 4°C overnight. The gel produced was washed, cut into pieces, and taken for assay.

The enzyme-entrapped gel was crosslinked[24] by treating with 2.5% solution of glutaraldehyde under stirring conditions for 24 h at 15°C. The preparation was washed and used for activity.

Immobilization of enzymes

Immobilization of enzymes by adsorption technique on different supports. In the preliminary studies on enzyme purification, it was found that cane amylase was adsorbed on TCP gel and other supports at pH 5.2 and was not eluted. Further experiments were carried out to optimize the conditions for insolubilization.

In a typical experiment, 200 mg each (dry basis) of supports as per list in *Table 2* were stirred in 0.1 M acetate buffer, pH 5.2, in a refrigerator for 4 h with 6 mg of enzyme protein in a total volume of 5.0 ml. The supports were removed by centrifuging at 4000 rev min^{-1} for 10 min in a refrigerated centrifuge, washed well with distilled water to remove free enzyme, and finally with 0.1 M acetate buffer, pH 5.2. Protein adsorbed was calculated by the difference of protein taken for immobilization and protein left in combined supernatant liquor and washings.

Assay of immobilized enzymes. (Amylase and acid phosphatase) A 5.0-ml reaction mixture containing 40–50 mg of (dry) treated supports per 100 beads per 1.2 g of polyacrylamide gel containing adsorbed/entrapped enzyme along with 2.5 ml 2% starch per 1 ml of 1% β-glycerophosphate in 0.1 M acetate buffer pH 5.4 was maintained at 50°C in a shaking incubator water bath for 1 h. The gel was removed by centrifugation.

Reducing sugar/inorganic phosphorus (p_i) released was estimated as above in the supernatant reaction mixture.

Effect of enzyme concentration and reuse of TCP gel. Optimum enzymes to carrier ratio on the adsorption/immobilization process was studied by varying the amounts of enzyme (1–7 mg protein) in 5 ml of 0.1 M acetate buffer, pH 5.2, with 200 mg of TCP gel (dry basis) and kept at 4°C under stirring conditions for 1 to 16 h. The (support) carrier at different intervals was removed by centrifuging and assayed for bound enzyme activity.

The support was removed by centrifuging in cold at 4°C after the reaction, washed with buffer, and checked again for activity as in the previous one by suspending in buffer with substrate. One hour reaction time of incubation of enzyme with substrate was taken as one cycle.

To study the properties of immobilized enzymes, the reaction mixture was incubated at different temperatures, pH, and different chemical agents. Thermal stability of the free and immobilized cane amylase and acid phosphatase was studied by treating with and without sucrose (18%) in 0.1 M acetate buffer, pH 5.4, at 65°C for different periods of time and then taken for enzyme assay.

Results and discussion

Isolation and purification of enzymes by salt and solvent fractionation and DEAE cellulose column chromatography showed the presence of both cane amylase and acid phosphatase in the fractions. Addition of NaF (10 mM) in the fractionation steps was not successful in the separation of enzymes. There was not much rise in specific activities in the purification steps (*Table 1*). The enzyme activities were overlapping in the eluates. Only the phosphocellulose column was able to separate cane amylase from acid phosphatase with the salt elution procedure of Abe *et al.*[16]

Sugar cane acid phosphatase was firmly adsorbed on phosphocellulose due to its great affinity to the substrate and was eluted only when the salt concentration was raised to 0.3 M.

Among the different supports taken for immobilization, TCP gel showed maximum binding of cane amylase and acid phosphatase (*Table 2*). Six milligrams enzyme protein on 200 mg TCP gel matrix in 4–6 h of contact time gave optimum results and the gel retained enzyme activity for about five cycles. Iron oxide beads showed amylase activity for two cycles and phosphatase activity for four cycles (*Figure 1*). TCP gel was found to be the best for enzyme immobilization, possibly due to its large number of hydroxy groups spread with water retention activity, high substrate affinity, complexing nature, and effective surface area for adsorption. Hydroxyl-apatite, a dehydrated form of TCP, was not found suitable for immobilization of these enzymes. Entrapment technique, although it retained enzyme protein, showed less activity. Similarly, Bentonite, Keiselguhr, Kaolin, Celite, etc., although they showed bound protein contents, produced negligible activities, possibly due to the active groups bound or modified in these operations.

The TCP-bound enzymes were quite stable and differed from glucoamylase on DEAE cellulose.[25,26] The enzymes did not show much leakage or desorption by treating with acetate buffer of 0.05 to 0.8 M, pH 5.2 (pH of buffers did not change with molarity used), and 0.1 M, pH 4 to 5.6, and 0.1 M tris maleate buffer of pH 7.0 (*Figure 2*). Glucoamylase on DEAE cellulose[25,26] and cane invertase on Bentonite[27] supports have been reported to suffer from leakage by changing the ionic strength (molarity) and pH of buffers used, as they are loosely attached to the supports.[25]

The pH and temperature optima for free and immobilized amylase were 5.2 to 5.5 and 57–60°C and for acid phosphatase pH 5.2 and 62–65°C temperature, respectively (*Figure 3*). There was not much alteration of these parameters in free and immobilized forms, showing that enzymes are immobilized on TCP gel

Table 1 Purification procedure for cane amylase and acid phosphatase from sugar cane juice

Steps	Protein (mg)	Activity (units)		Specific activities	
		Amylase	Acid phosphatase	Amylase	Acid phosphatase
Crude enzyme	630	6630	100.8	10	0.16
Ammonium sulfate fraction (80% saturation)	390	6630	70.2	17	0.18
NaF treatment in acetate buffer and acetone fraction (50% saturation)	300	5400	60.0	18	0.20
DEAE cellulose	226	4524	49.7	20	0.22
Phosphocellulose (0.15 M NaCl)	40	1120	0	28	0
(0.30 M NaCl)	21	0	5.7	0	0.27

Table 2 Immobilization of cane amylase and acid phosphatase on different supports[a] by entrapment and adsorption technique

Support		Protein retained (mg)	Specific activity		Specific activity retained (percent)		Cycle time	
			Amylase	Phosphatase	Amylase	Phosphatase	Amylase	Phosphatase
Polyacrylamide gel	Entrapment	6.0	8.0	0.08	47.1	44.4	1	1
Polyacrylamide gel (crosslinked)	Entrapment	6.0	7.5	0.08	44.1	44.4	1	1
Beads (calcium alginate)	Entrapment	6.0	4.3	0.04	25.3	22.3	1	1
Beads (iron oxide)	Entrapment	6.0	7.5	0.08	44.1	44.4	2	4
Beads (magnetite)	Entrapment	6.0	6.2	0.05	36.5	27.8	1	2
Bentonite	Adsorption	5.0	0.2	0.08	1.2	44.4	N.D.	N.D.
Kaolin	Adsorption	3.8	0.3	0.04	1.8	22.2	N.D.	N.D.
Kieselguhr	Adsorption	4.1	0.1	0.03	0.6	16.7	N.D.	N.D.
Celite	Adsorption	4.0	0.1	0.04	0.6	22.2	N.D.	N.D.
Hydroxyl apatite	Adsorption	4.1	8.2	0.08	48.2	44.4	N.D.	N.D.
Tricalcium Phosphate (TCP) Gel protein added (mg)	Adsorption							
1	Adsorption	1.0	12.2	0.13	71.8	72.2	1	2
2	Adsorption	1.9	13.7	0.14	80.6	77.8	2	3
4	Adsorption	3.8	13.4	0.14	78.8	77.8	3	3
5	Adsorption	4.5	14.2	0.14	83.5	77.8	5	5
6	Adsorption	5.4	14.4	0.15	84.7	83.3	5	5
7	Adsorption	5.1	14.3	0.15	84.1	83.3	4	5

[a] Protein taken 6 mg. specific activity: [amylase = 17 units mg^{-1}; acid phosphatase = 0.18 units mg^{-1}]
N.D., Not determined

without any appreciable modification of active groups.

The thermal stability of cane amylase and acid phosphatase in free form was found to be 44% and 65% respectively, as compared to initial values on heating at 65°C for 60 min. The heat stability in the immobilized forms of these enzymes was found to be 83% and 92% respectively, as compared to the initial value. In the presence of sucrose (18%), both free and immobilized forms of cane amylase and acid phosphatase on heating at 65°C for 60 min showed high thermal tolerance, showing 73% and 81% of initial value in free form, and in the immobilized form the activities were 95% and 100%, respectively (*Figure 4*).

The rise in heat tolerance is due to the protective nature of support matrices. Sucrose, due to its high polar nature, protected the enzyme from thermal denaturation; similar results were observed with dextranase immobilized on bentonite.[28] Cane acid phosphatase with differing K_m and V_{max} values using pyrophosphate and β-glycerophosphate as substrate gives an indication of kinetic differentiation (*Figure 5*). Similar data on acid phosphatase have been reported by Sampietro *et al.*[29] in sugar cane stalk with β-glycerophosphate and p-NPP as substrate.

It can thus be concluded that phosphatase present in cane juice may not be specific and can break down

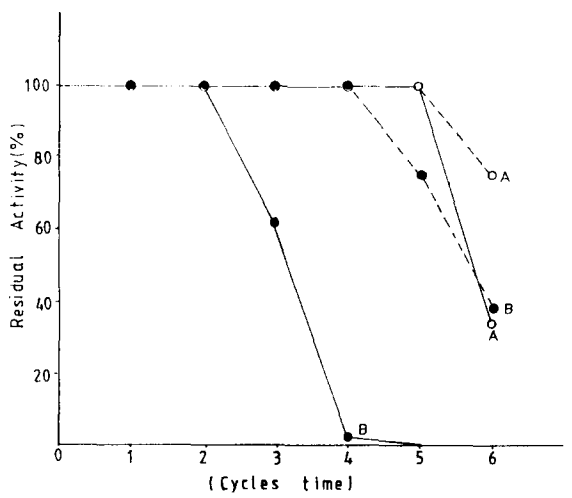

Figure 1 Repeated use of immobilized cane amylase (—) and phosphatase (----). A, TCP gel; B, beads

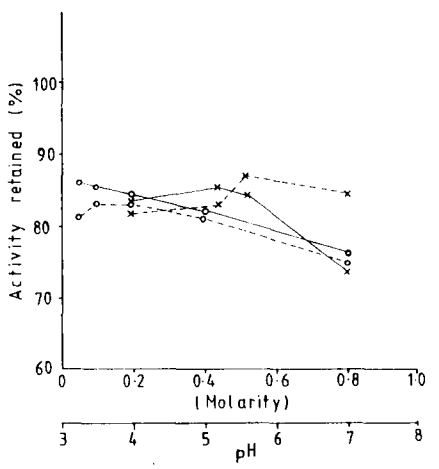

Figure 2 Effect of molar concentration (O) and pH of buffers (X) on binding of cane amylase (—) and acid phosphatase (----)

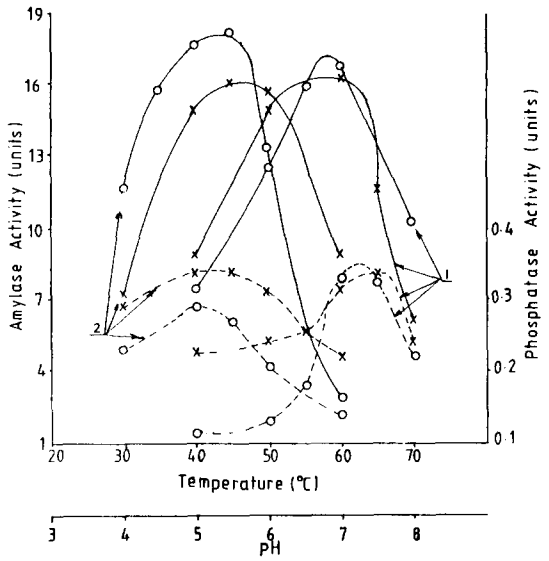

Figure 3 Effect of temperature (1) and pH (2) on free (○) and immobilized (X) cane amylase (—) and acid phosphatase (----)

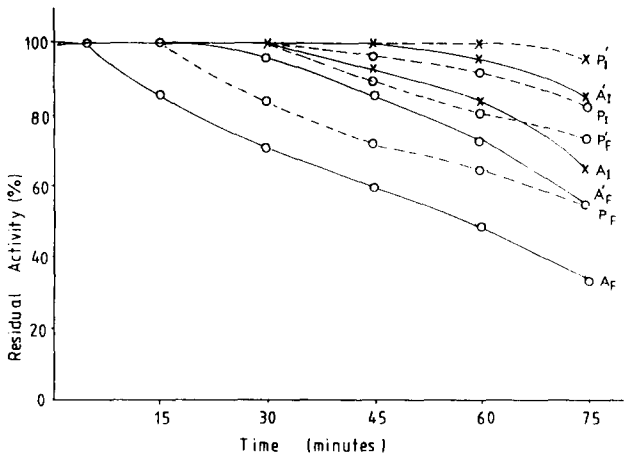

Figure 4 Thermal stability of free (○) and immobilized (X) amylase (—) and acid phosphatase (----) from sugar cane at 65°C ± sucrose (18%). A_F, A_I, Amylase free and immobilized; A'_F, A'_I, amylase free and immobilized + sucrose; P_F, P_I, acid phosphatase free and immobilized; P'_F, P'_I, acid phosphatase free and immobilized + sucrose

different substrates at different rates. Lower values of K_m for pyrophosphate as compared to β-glycerophosphate showed that cane meets its requirements of phosphates via breaking the inorganic phosphate complex rather than β-glycerophosphate,[9] a reservoir of important organic phosphate compounds in the sugar cane plant.

The increase in K_m values of acid phosphatase and cane amylase on immobilization with or without much alteration in V_{max} as compared to its free form (*Figures 5* and *6*) suggested that there is a significant decrease in the affinity of the enzymes for their substrates, and the enzyme protein is immobilized not only on the surface but also in TCP particles of gel. K_m depends on the formation of the ES complex, and V_{max} is a measure of the breakdown of the latter.[30] Similar results have been reported earlier for immobilization of glucoamylase on DEAE cellulose[26] and dextranase on bentonite.[28]

SH group binding agents such as p-CMB and $CuSO_4$ showed complete inhibition of cane amylase and much less of acid phosphatase. Similarly, surface active agents, viz. SLS and CTPB, strongly inhibited cane amylase and to a lesser extent acid phosphatase. The inhibitory action of compounds was more in the free form in acid phosphatase as compared to the immobilized form, except in the case of $CuSO_4$ where it was nearly equal in both the forms (*Table 3*).

Cane amylase completely differed from glucoamylase with respect to SH group inhibition.[26] NaCN, α-α^1-dipyridyl had no effect on both the enzymes, indicating that no metal ion is involved as cofactor, as reported in bacterial amylase,[31] and pancreatic amylase,[32] Ca^{+2}, Mn^{+2}, and NaN_3 did not show inhibition of cane amylase and acid phosphatase. Sodium metasilicate inhibited both the enzymes in free and immobilized forms; similar results have been reported by Alexander.[33]

The ease and economy of active immobilized preparations of enzymes indicate their usefulness in indus-

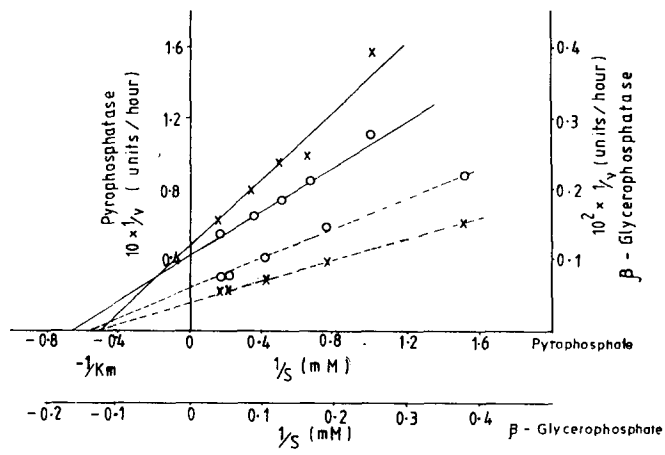

Figure 5 Lineweaver–Burk plots for acid phosphatase. Free pyrophosphate (○—○); free β-glycerophosphate (○----○); immobilized pyrophosphate (X—X); immobilized β-glycerophosphate (X----X)

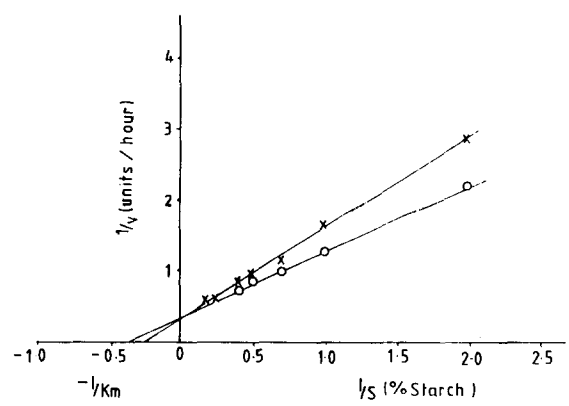

Figure 6 Lineweaver–Burk plots for free (○) and immobilized (X) sugar cane amylase

Table 3 Effect of different chemicals on free and immobilized cane amylase and acid phosphatase

Sl.	Chemicals (10 mM)	Relative activity (%)			
		Amylase		Phosphatase	
		Free	Immobilized	Free	Immobilized
1	Control	100.0	100.0	100.0	100.0
2	$ZnSO_4$	90.2	68.0	50.0	58.0
3	$CuSO_4 \cdot 5H_2O$	0	0	50.0	49.0
4	$CoCl_2$	110.0	110.0	100.0	100.0
5	$CaCl_2$	100.0	100.0	100.0	100.0
6	$MgSO_4$	106.0	94.0	66.6	71.0
7	$MnSO_4 \cdot H_2O$	110.0	103.0	100.0	100.0
8	Sodium meta silicate	20.0	26.0	4.0	20.0
9	NaCN	100.0	100.0	100.0	100.0
10	NaN_3	100.0	100.0	100.0	100.0
11	NaF	100.0	140.0	46.3	72.7
12	P-CMB (p-chloromercuribenzoate)	0	0	68.8	100.0
13	α-α^1-dipyridyl	95.9	100.0	105.0	100.0
14	Sodium laurylsulfate (SLS)	14.5	14.9	39.2	78.0
15	Cetyl pyridinium bromide (CTPB)	10.0	15.0	75.0	80.0

try. The application of this TCP gel bound enzyme system of cane amylase and acid phosphatase in sucrose solution or cane juice will greatly help in continuous removal of starch.

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References

1 Martensson, K. *Biotechnol. Bioeng.* 1974, **16,** 567, 579
2 Hultin, H. O., Laurence, R. L. and Kittrell, J. R. *Enzyme Technol Dig.* 1974, **2,** 156
3 Katwa, L. C. and Rao, M. R. R. *Biotechnol Lett.* 1983, **5,** 191–6
4 Obata, H., Tanishita, J., Tokuyama, T. *Technol. Rep. Kan Sai Univ.* 1983, **24,** 193–198
5 Alexander, A. G. *J. Agric Univ. PR,* 1965, **49,** 153–175
6 Alexander, A. G. and Lebran, J. *International Society of Sugarcane Technologists,* 13th Congress, 1968, 514–521
7 Vignes, E. C., Abel, M. and Guen, L. *Mauritius Sugar Ind. Res. Inst. Ann Rpt.* 1969, 141–145
8 Nicholson, R. I. and Horsley, M. *Int. Sugar J.* 1958, **60,** 260–263
9 Das, G. and Prabhu, K. A. *Int. Sugar J.* 1988, **90,** 67–71
10 Das, G. and Prabhu, K. A. *51st Proc. Sugar Technologists Association of India* 1988 (in press)
11 Meade, G. P. and Chen, J.C.P. *Cane Sugar Handbook,* 10th ed., John Wiley & Sons, New York, 1977, p. 183
12 Keilin, D. and Hartrce, E. F. *Proc. Roy Soc. (London)* 1938, **B129,** 397
13 Jenkins, Terry W. in *Biochemical Preparations* (Coon, M. J., ed.) Vol. 9, John Wiley and Sons, New York, London, 1962, p. 83
14 Burns, M. A., Kvesladze, G. I. and Graves, D. J. *Biotechnol Bioeng.* 1985, **27,** 137–145
15 Fleming, I. D. and Stone B. A. *Biochem. J.* 1965, **13,** 97
16 Abe, J., Takeda, Y., Hizukuri, S. and Hasegawa, Y. *Biochim. Biophys. Acta* 1982, **703,** 26–33
17 Venkataramu, K., Manjunath, P. and Rao, M.R.R. *Ind. J. Biochem. Biophys.* 1975, **12,** 107
18 Colowick, S. P. and Kaplan, N. P. *Methods Enzymol.* 1955, **2,** 523
19 Miller, G. L. *Anal. Chem.* 1959, **31,** 426
20 Fiske, C. H. and Subbarow, Y. *J. Biol. Chem.* 1925, **66,** 375
21 Lowry, O. H., Rosebrough, N. J., Farr, A. L. and Randall, R. J. *J. Biol. Chem.* 1951, **193,** 265
22 Brodelius, P., Dens, B., Mosbach, K. and Zenic, M. H. *FEBS* Lett. 1979, **103,** 93
23 *Immobilized Enzymes* Ichiro, C., ed.) John Wiley & Sons, New York, 1978, p. 52
24 Chang, T. M. S. *Biochim. Biophys. Res. Comm.* 1971, **44,** 1531–1536
25 Bachler, M. J., Strandberg, G. W. and Smiley, K. J. *Biotechnol. Bioeng.* 1970, **12,** 85
26 Mamta, T. and Prabhu, K. A. *Enzyme Microb. Technol.* 1985, **7,** 557–560
27 Mamta, T. and Prabhu, K. A. *Enzyme Microb. Technol. 1985,* **7,** 454–458
28 Madhu and Prabhu, K. A. *Enzyme Microb. Technol.* 1985, **7,** 279–282
29 Sampietro, A. R., Etcheberrigaray, J. L., Quiroga, E. and Arias de Flores, E. *Rev. Ind. Agric. Tucuman* 1976, **53,** 17–23
30 Montgomery, R. and Swenson, A. C. *Quantitative Problems in the Biochemical Sciences* W. A. Freeman, San Francisco, 1976, p. 222
31 Rischer, E. H. and Stein, E. A. in *The Enzymes,* 2nd ed., Vol. IV (Boyer, P. D., Lardy, H. and Myrback, K., eds.) Academic Press, New York, 1960, p. 313
32 Cold-Weil, M. L., Weil, C. E. and Weil, R. S. *J. Am. Chem. Soc.* 1945, **67,** 1079
33 Alexander, A. G. in *Sugarcane Physiology* Elsevier Scientific Publishing Company, Amsterdam-London, New York, 1973, p. 472–474